COMPETENCE BASED BIOS QUESTIONS

CBBQ



QUESTIONS 2025

Specific Competence Assessment Tools

FORM ONE 2025 SCAT- 1

F1 -MIDTERM-1 EXAMS QUESTIONS.

The basic concepts of Biology

1. Use the knowledge of branches of biology to attempt the following briefly
   a. Name the branch of biology involved in each of the following
      i. Study of inheritance and variations
      ii. Study of cells
      iii. Study of parasites
      iv. Science field that specializes in the study of chemical changes in an organism.

   b. Describe what each of the following branches of biology involves

      i. Haematology
      ii. Entomology
      iii. Microbiology

   c. Complete the table below appropriately

Branch of biology	*Name of scientist who specializes in it.*	*Name of study*
Zoology		
	Ecologist	
		Study of fungi

2. Use brief notes from the study of life processes to
   i. Explain use one point in each how does nutrition as living things characteristic differ in plants, animals, fungi and viruses.
   ii. State the characteristics of living things that is being demonstrated by plants producing carbon (IV) oxide during daytime.

iii. Explain this biological phenomenon “Gaseous exchange and cellular respiration are characteristics of living things and are said to be agonistic processes in the living organism”
iv. Suggest with three reasons why animals move from one habitat to another.
v. Explain this observation use two points “A car produces energy, moves about and produces waste products yet it is regarded as a nonliving thing”.
vi. Explain why is it necessary to expand biology to include the study of non-living things.
vii. Name the characteristics of living organisms, illustrated through the following activities.
    a. A cow giving birth to a calf.
    b. A student eating “Githeri” in the dining hall.
    c. A gazelle running away after spotting Lion from a distance.
    d. A student sneezing after smelling strong perfume.
    e. Amoeba remove there metabolic by product in the body by thin layer ectoplasm

3. Attempt the following from the study of biological branches, field and characteristics of living things to,
    i. Name the main branch of Biology that studies the following.
        a. Organisms such as a Crocodile.
        b. Organisms seen as a bean plant.
    ii. Explain the following biology sub-branches
        a. Mycology
        b. Ornithology
        c. Anatomy
        d. Cytology
        e. Bio chemistry
    iii. Explain the following characteristics of living organisms
        a. Growth
        b. Development
    iv. Differentiate how does growth as a characteristic of living organisms differ in plants and animals
    v. State any two environmental problems that the study of biology tries to solve.

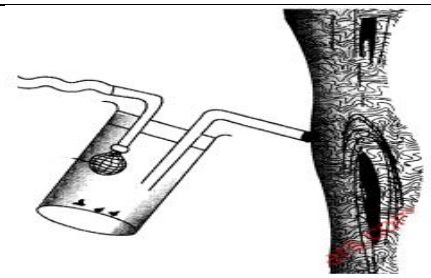
4. Using your knowledge of basic concept and terminology to attempt the following
    a. Biology is derived from two Greek words bios and logos. What is the meaning of
    Bios……..
    Logos ……..
    b. Name the branch of Biology that deals with the study of animals.
    c. State two importance of the following life processes in living organisms

i. Nutrition
ii. Excretion

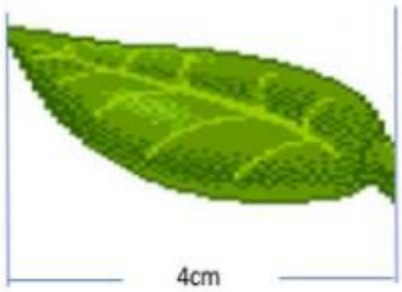
d. Explain four reasons why you would encourage your friends in High school to study Biology

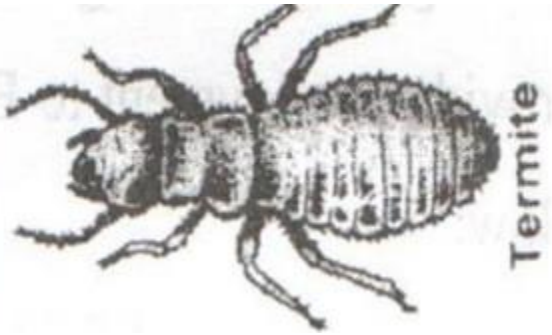
Simple biological experiment

5. Use your comprehension knowledge in biological experiment to
   a. State any three scientific skills gained through the study of biology.
   b. State four precautions during collection and observation of specimen.
   c. State the suitable apparatus when some form one students wanted to collect and study the following specimen in the biology laboratory.
      i. Crawling stinging insects
      ii. Safari ants on a tree trunk
      iii. Scorpion
   d. Below is an illustration of a pieces of apparatus strategically positioned to trap some organism state the

NAME	APPARATUS	FUNCTION	ORGANISM EXAMPLE
			
			
			

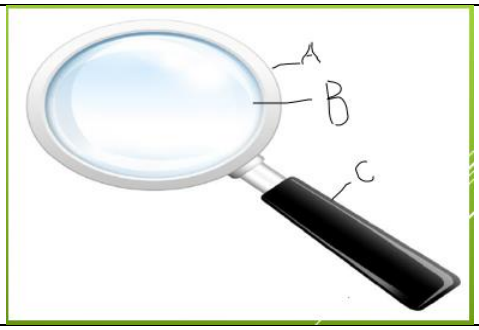
6. Apply your knowledge of methods of studying magnification to:
   a. State the meaning of magnification
   b. Calculate the magnification of the drawing of the termite below given that the actual length of the termite is 0.9cm long. Show your working clearly.
   c. The following type of leaf below were from botanist drawing of certain biological field; he uses magnification power of x10.
      i. Calculate the actual length of the leaf (Show your working clearly).
      ii. Name this type of leaf shape.





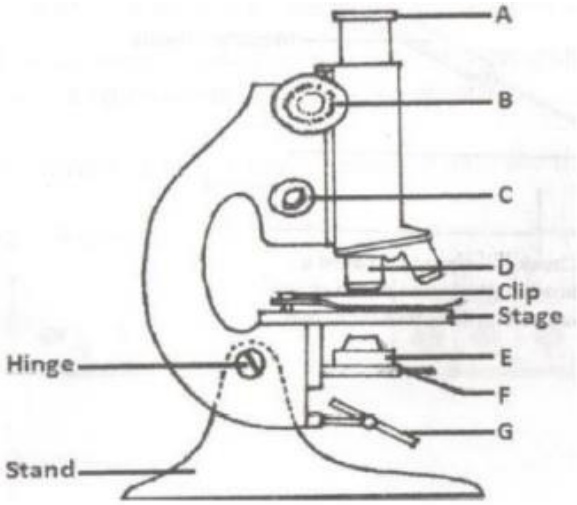
d. A student observing a leg of an insect under a hand lens made a drawing of the leg whose length was 4cm with a magnification of X2. What was the actual length of the insect leg.?

7. From the study of simple biological apparatus apply the following.
   a. Complete the following table

NO	NAME	APPARATUS	FUNCTION
1.			
2.			

   b. Label part A, B and C of apparatus (2)
   c. New – born babies have generally a 35 number of beats per 15sec while that of adult is 17.5 number of beats per 15sec.
      i. How many heart – beat count per minute for new born babies and adult.
      ii. Account for the difference.
   d. New – born babies have generally a heart – beat count to 120 to 140 per minute while that of adult is 70 per minute on average. Account for the difference
   e. Convert the following quantities
      i. 90 degrees of Fahrenheit to degree of centigrade.
      ii. 40 degrees of centigrade to Fahrenheit.

8. The diagram below represents the structure of a light microscope, study it and answer the questions



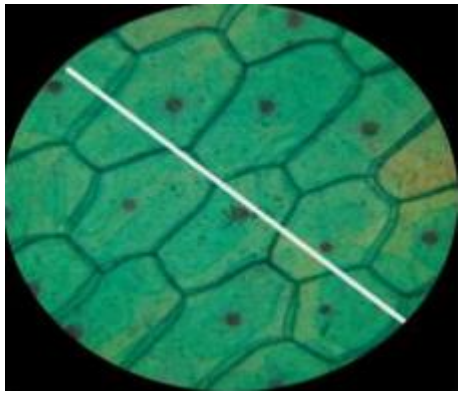
   a. Name the parts of the microscope labeled A to G
   b. State the functions of each of the part marked E and F
   c. Which part of the microscope
      i. Contribute to the magnification of the specimen
      ii. Is used to move the body tube over very small distance when observing under high power magnification
   d. Using arrows show how the object is illuminated.

9. Use your knowledge of methods of studying microscope to;
   a. Suggest two possible causes of this**, "while trying to observe a specimen under a microscope, student failed to see the field of view".**
   b. Explain why the following are important during the preparation of temporary slides under microscopic specimen.
      i. Cutting very thin sections
      ii. Using a sharp razor blade during the cutting
      iii. Placing sections in water
      iv. Staining the sections with iodine before observing
      v. Placing cover slip over the plant section
   c. Give a reason under which of the following magnification would one see a larger part of the specimen X40 or X500?
   d. A student viewing a prepared slide of a plant cell under high power microscope. The features of the cell were blurred. Name the parts of the microscope would the student will obtain:
      i. A sharper outline of the features
      ii. Less sharp outline of the features

10. Apply your knowledge of methods of studying magnification in light microscope to;

a. Explain the meaning of resolving power

b. Use the following diagram shows onions cells captured in a field of view of a light microscope to,



i. Measure the length of the white line to determine the diameter of the field of view in millimeters.

ii. How many cells are found along the diameter represented by the white line?

iii. Determine the actual diameter of one cell in micrometer if a magnification of X1000 was used to observe the cells above

c. During a practical class, form one estimated the field of view to be 3.5mm. Using the low power objective, they observed spirogyra cells across the same field of view and counted 8cells. Calculate the size of each cell and give your answer in micrometer.

d. In a laboratory exercise a student observing a drop of pond water under a microscope saw and drew a spirogyra. If the magnification of the eye-piece was x5 and that of the objective lens was x100.

i. what was the magnification of the spirogyra?

ii. If the spirogyra has a length of 5cm at the above magnification, calculate the actual length in micrometers.

iii. Give a reason why an electron microscope cannot be used to study living specimen organism.

Describe a cell

11. Use the concept from cell anatomy to;

a. Explain the term cell

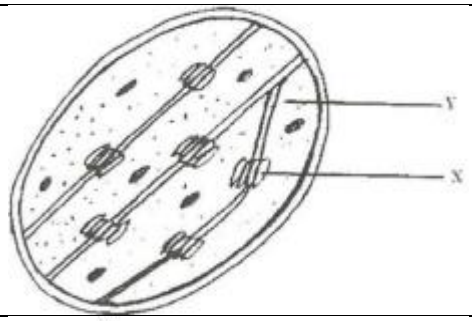
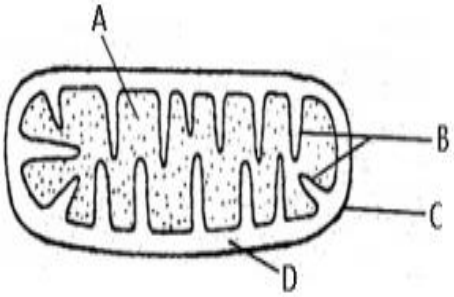
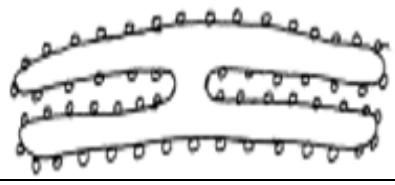
b. Identify the type of the cell base on the following statement “. If a cell was found to have the following under a light microscope; cell membrane, irregular in shape, and small vacuoles. “

c. Name two structures present in plant cell but absent in the animal cell

d. Identify the cell organelle and state the appropriate function

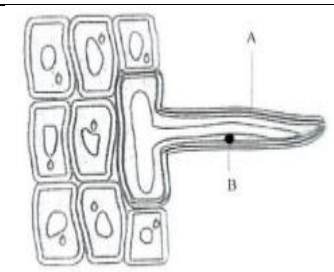
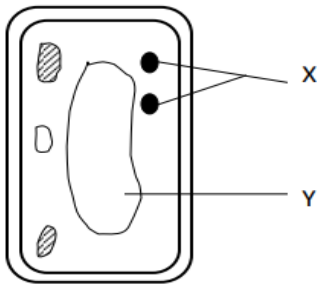
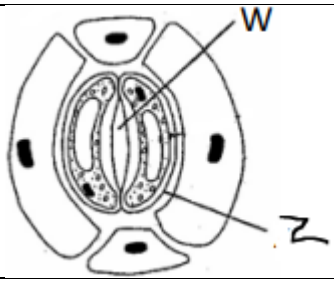
CELL ORGANELLE	FUNCTION
Cell wall	.....
.....	Add carbohydrates to protein and transport them in the cell
Nucleus	.....
Nucleolus	.....
.....	Protein synthesis
Chloroplast	.....
.....	Contain lytic enzymes
Rough endoplasmic reticul	.....
.....	Transport lipids
.....	Site for respiration

12. Use the comprehension knowledge of cell organelles to answer the following briefly.

NO	NAME	ORGANELLES	FUNCTION
1.			
2.			
3.			

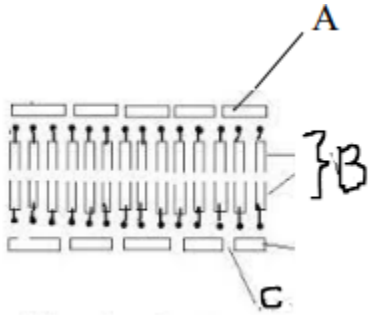
a. Complete the above table
b. From above organelle (1) and (2), which one is abundant in the following cell and state briefly why?
    i. Muscle
    ii. Palisade
c. What is the importance of infoldings in the inner membrane in cell organelle (2).
d. Give two examples of tissues where you would expect find many of organelles in cell organelle (2) in animal body.
e. Name the organelle that,
    i. Manufacture and transport lipids and steroids in a cell
    ii. Contain enzymes that are capable of destroying old damaged cells.
    iii. Control all the processes in a cell.
    iv. Form Cilia and flagella in cells
f. Label parts X, Y, A, B, C and D

13. Use the concept of cell specialization to attempt the following

NAME	DIAGRAM	FUNCTION
		
		
		

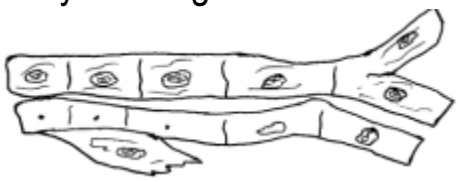
a. Complete the above table
b. Label the following parts A, B, X, Y, W and Z
c. State what make each of above cells to be specialized cell
d. State two adaptations of cell labelled Z to its function.
e. Guard cells are specialized epidermal cells. State two structural features which suit them to their functions.

14. From the concept of cell physiology provide briefly notes from the diagram.



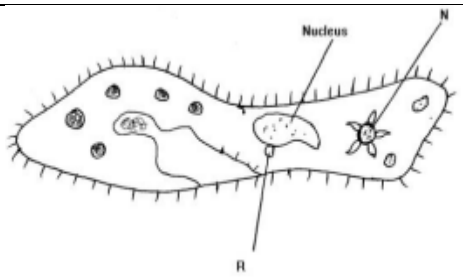
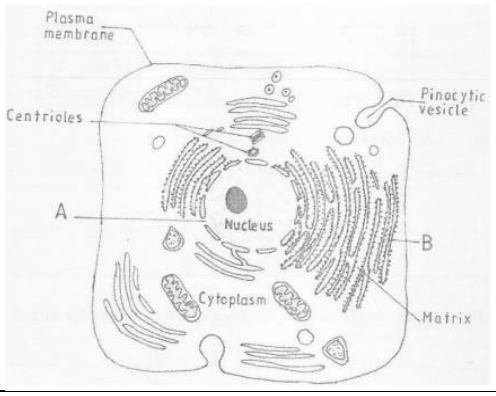
a. Label the following parts A, B and C
b. State two properties of the structure above
c. Mention the property that regulate exchange of substances in and out of the nucleus
d. Name the others two principal components of cell membrane
e. Why would a cell allow some substances to pass through it but not others?

15. Study the diagram below and answer the questions that follow.



a. What is the identity of the muscle tissue shown.
b. Where is the tissue found
c. What make it to be specialized tissue
d. Give one special property of the muscle which is not common to other muscle.

16. From the concept of type of cell briefly answer the following

NO	NAME	CELL	CELL TYPE
1.			
2.			

a. Completely the above table.
b. How is the above cell differing. Give four points.
c. Is this observation under a light or an electron microscope? Give a reason for your answer.
d. Is cell (2) an animal or a plant cell? Give a reason for your answer.
e. Labell the parts N, R, A and B

FORM ONE PRACTICAL

AIM: To investigate the fine structure of a generalized onion bulb cell as seen under light microscope.

Apparatus

- *Onion bulb*
- *Methyl blue (5ml) per group*
- *Cover slip (1pc)*
- *Distilled water (liquid H)*
- *Blotting paper/cotton wool*
- *Scalpel blade*
- *A pair of forceps*
- *Microscope (one for a group of five)*
  N/B – You are provided with the following objective lens x5, x10 and x45 and eye piece lens of x10

Procedures

- You are provided with a portion of an onion bulb.
- Remove one fleshy leaf from the onion bulb.
- Using scalpel blade to peel the epidermis from the inner surface of the leaf.
- Use pair of forceps to pick the cut leaf and place it on a drop of methyl blue on a glass slide.
- Place a cover slip on the epidermis.
- Drain the excess methyl blue by use of a piece of blotting paper/cotton wool from the edge of the cover slip then leave the set up for **1 minutes**.
- Place a drop of **liquid H** at the edge of the cover slip.
- Leave the set up for **5 minutes** then drain excess liquid from the opposite of the slip using a blotting paper/cotton wool.
- Observe under low power, medium power and high power of the light microscope provided.

Questions

(a) **Draw** and **label** four neighboring cells observed at x45.

(b) Account for the results in **(a)** above

(c) What was the function of the following processes during the preparation of temporary slides.

i. **Cutting very thin sections**
ii. **Using a sharp razor blade during the cutting**
iii. **Placing sections in liquid H**
iv. **Staining the sections with methyl blue before observing**
v. **Placing cover slip over the plant section**

(d) Estimate the average size of each cell in micrometer, assuming diameter of the field of view is 4mm and 6mm in x10 and x45 magnification power respectively. Show your working clearly.

Given that:

$$Diameter\ of\ one\ cell = \frac{diameter\ of\ field\ of\ view}{number\ of\ epidemal\ cell}$$

(e) State one limitation of the method above for illustrating cell size.

(f) What is the total magnification of the image of an onion epidermal cell per each objective lens.

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Prefer Calling Group of Examiners @ 0738380920 for Marking Schemes and the rest of semester question 2025

